

ATPL EASA STUDY GUIDE | 081 - PRINCIPLES OF FLIGHT

Chapter 13: Flight Controls, Control Tabs & Aerodynamic Balance

Syllabus Source: CAE Oxford (pp. 465-504) | Pass Target: >= 90% AviationExam

1. PRIMARY CONTROLS & ADVERSE YAW MITIGATION

Primary flight controls govern rotation about the aircraft's three axes (Ailerons: Roll; Elevator: Pitch; Rudder: Yaw):

- **Adverse Yaw Phenomenon:** When ailerons deflect to initiate a roll, the downward-deflected aileron on the rising wing produces more lift AND MORE INDUCED DRAG than the upward-deflected aileron on the falling wing. This differential drag yaws the nose in the direction OPPOSITE to the commanded bank!
- **Differential Ailerons:** The up-going aileron deflects through a substantially LARGER angle (e.g. 25° up) than the down-going aileron (e.g. 10° down). The large up-deflection generates extra parasite form drag on the inside wing, counteracting adverse yaw.
- **Frise Ailerons:** The aileron hinge is set back from its leading edge. When deflected UP, the lower lip of the aileron protrudes below the wing lower surface into the slipstream, creating parasite drag on the down-going wing to balance adverse yaw.
- **Roll Spoilers:** Spoilers deflect upward ONLY on the down-going wing, destroying lift and adding large parasite drag on the inside of the turn, completely eliminating adverse yaw at high speeds.

2. CONTROL TABS & AERODYNAMIC BALANCING

Aerodynamic balancing reduces the physical hinge moment (stick force) required from the pilot:

Control Tab / Balance Device	Mechanical Deflection Direction	Aerodynamic Function & Stick Force Effect
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Trim Tab (Controllable)	Deflects OPPOSITE to primary control surface (moved by cockpit trim wheel).	Holds control surface in deflected position with ZERO stick force from pilot. Trims aircraft for hands-off flight.
Balance Tab	Mechanically geared to deflect OPPOSITE to primary surface automatically.	Reduces pilot control stick forces by using tab aerodynamic lift to help deflect main surface.
Anti-Balance Tab	Mechanically geared to deflect in the SAME DIRECTION as primary surface.	INCREASES control feel and stick forces! Mandatory on all-flying stabilators (Piper Cherokee) to prevent pilot over-controlling and overstressing airframe.
Servo Tab	Directly connected to pilot cockpit controls; main surface is free-floating.	Pilot moves tab only; tab aerodynamic force pushes the entire large control surface into position (BAe 146, DC-9).
Spring Tab	Incorporates a torsion spring in the linkage.	At low speed, spring acts as rigid rod (direct manual control). At high dynamic pressure, spring compresses, allowing tab to assist pilot as a servo tab.

[!] EXAM TRAP (AVIATIONEXAM) - Balance Tab vs Anti-Balance Tab Direction Trap

- Balance Tab: Moves OPPOSITE to control surface -> REDUCES control stick forces! -
 Anti-Balance Tab: Moves in SAME DIRECTION as control surface -> INCREASES control stick forces and artificial feel! - Trim Tab: Cockpit trim nose-down commands trim tab UP, which aerodynamically forces the elevator DOWN.

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)**[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Flight Control Tabs Deflection Sequence**

Problem: You trim elevator nose-up (primary surface UP). Which way does the trim tab deflect? Step 1: A conventional trim tab generates aerodynamic force to hold the trailing edge of the primary control. Step 2: To keep the elevator UP, the tab must generate a downward aerodynamic force -> Tab deflects DOWN. Exam Rule: Trim tab and Balance tab move in the OPPOSITE direction of the control surface. Anti-balance tab moves in the SAME direction.

3. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)**Q1: How do DIFFERENTIAL AILERONS reduce adverse yaw during a roll maneuver?**

- [A]: By disconnecting the rudder
- [B]: The up-going aileron deflects through a GREATER angle than the down-going aileron, producing extra parasite drag on the inside wing to counter adverse induced drag
- [C]: By lowering the flaps
- [D]: By moving both ailerons downward together

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. The up-aileron deflects significantly farther than the down-aileron, creating deliberate form drag on the descending wing to counteract the adverse induced drag of the rising wing.

Q2: In what direction does an ANTI-BALANCE TAB move relative to the main control surface, and what is its purpose?

- [A]: In the opposite direction, to reduce control forces to zero
- [B]: In the SAME DIRECTION as the control surface, to INCREASE stick forces and provide artificial aerodynamic feel on an all-flying tail
- [C]: It remains locked in cruise
- [D]: It trims the rudder only

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. An anti-balance tab deflects in the same direction as the surface, opposing pilot input and increasing stick force per g, preventing over-controlling on light all-flying stabilizers.

Q3: When the pilot trims for NOSE-UP pitch using a conventional elevator trim tab, in which direction does the trim tab deflect?

- [A]: The trim tab deflects UP
- [B]: The trim tab deflects DOWNWARDS, creating an upward aerodynamic force that pushes and holds the elevator UP
- [C]: The tab does not move

- [D]: The tab moves laterally

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. To hold the elevator up for nose-up trim, the trim tab at the elevator trailing edge must deflect DOWN, generating upward aerodynamic lift that forces the elevator trailing edge up.